

Lab 3 — Factorization and Embeddings in Practice

Machine Learning II · Tutorial

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Wednesday, October 21, 2026 · 19:00–22:00

Goal of the session. Sections 1 and 2 of Homework 3: the SVD and factorization problems on paper, matrix factorization by SGD on a panel with missing entries, and word2vec trained from scratch on a small corpus — with the gradients derived before the code runs.

Materials

- `homework_3_student.pdf` / `homework_3_solutions.pdf` — Sections 1 and 2.
- `homework_3_student.ipynb` / `homework_3_solution.ipynb` — Parts 1 and 3.
- `data/dr_seuss.txt` — the corpus for word2vec.
- Lectures 4 and 5 of ML II (latent factors, ALS vs. SGD, NMF; skip-gram, negative sampling).

1 SVD and Factorization on Paper (40 min)

Task 1 — Section 1 of the homework

Work Problems 1–6 in order. The ones to be sure of before the code:

1. $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ and its extension to three factors (Problem 1).
2. $\mathbf{X}^\top \mathbf{X} = \mathbf{VD}^2 \mathbf{V}^\top$ from the SVD, and why $\mathbf{X}^\top \mathbf{X}$ is positive semidefinite (Problem 2) — ML I, Lecture 6.
3. The storage count of a rank- k truncation, $k(2N + 1)$, and when it beats N^2 (Problem 3).
4. The SGD gradients of $\frac{1}{2} \sum_{(i,j) \in \Omega} (Y_{ij} - \mathbf{u}_i^\top \mathbf{v}_j)^2 + \frac{\lambda}{2} (\|\mathbf{U}\|_F^2 + \|\mathbf{V}\|_F^2)$ with respect to $\mathbf{u}_i$ and $\mathbf{v}_j$ (Problem 4). You will implement exactly these.
5. The ALS closed form for one row with $\lambda = 0$, $K = 1$, and the counterfactual $\hat{\tau}_{3,2} = 2$ (Problem 6).

2 Matrix Factorization by SGD (50 min)

Task 2 — Part 1 of the notebook, Problem 17(a)–(b)

The notebook generates a 20×12 panel (firms $\times$ months) of log sales with known rank-2 structure plus noise, and masks 25% of the entries as a staggered treatment. Implement the SGD updates from Problem 4 with $\mathbf{U}, \mathbf{V}$ initialized uniformly in $[-0.5, 0.5]$ and $\eta = 0.03$.

Sweep $K \in \{1, 2, 3, 5, 8, 12\}$ and plot in-sample error E_{in} (observed entries) and held-out error E_{out} (masked entries).

Task 3 — what the curves say

E_{in} falls monotonically in K ; E_{out} turns up. Find the K that minimizes E_{out} and compare with the true rank. This is the bias–variance trade-off of ML I, Lecture 1, with K as the capacity knob and Eckart–Young (ML I, Lecture 6) saying what happens at $K = 12$ with no missing entries.

Task 4 — ALS for comparison

Implement one ALS sweep (solve for every $\mathbf{u}_i$ with $\mathbf{V}$ fixed, then every $\mathbf{v}_j$ with $\mathbf{U}$ fixed; each is a ridge regression — ML I, Lecture 3). Compare with SGD at the same K : iterations to converge, final E_{out} , wall-clock time.

3 Word2vec From Scratch (50 min)

Task 5 — the gradient first (Section 2, Problem 10)

Using $\frac{d}{dz}[-\log \sigma(z)] = \sigma(z) - 1$, derive the negative-sampling gradient with respect to the centre-word vector and to one negative sample’s vector. Count the cost per update: $O((k+1)D)$ against $O(WD)$ for the full softmax.

Task 6 — Part 3 of the notebook, Problem 19

1. Tokenize `dr_seuss.txt`; generate (centre, context) pairs with window 2.
2. Train the skip-gram network with negative sampling in torch, $D = 16$, 5 negatives, 20 epochs.
3. Extract the embedding matrix; list the 30 most similar word pairs by cosine similarity.
4. Reflection: which pairs make sense, which are artefacts of a 1,500-word corpus, and what would change with a corpus 10^4 times larger.

Task 7 — word2vec is matrix factorization

Lecture 5 stated that skip-gram with negative sampling implicitly factorizes the shifted PMI matrix (Levy & Goldberg, 2014). Build the word–context co-occurrence matrix for the corpus, compute $\text{PMI} - \log k$, take its rank-16 truncated SVD (ML I, Lecture 6), and compare the nearest-neighbour lists with those of Task 6. They will not be identical; note where they agree.

4 Exercises (to submit before Lab 4)

Exercise 1. Section 2, Problems 7–9 (LLE reconstruction weights and the hierarchical-softmax path length).

Exercise 2. Regularization sweep for the factorization, Problem 17(d): $\lambda \in \{0, 0.01, 0.1, 1\}$ at the best K and at $K = 12$. Explain why λ matters more at large K using the ridge intuition from ML I.

Submission

Notebook `lab03_lastname_firstname.ipynb` and a PDF of the paper work, before Monday, October 26, 18:00.